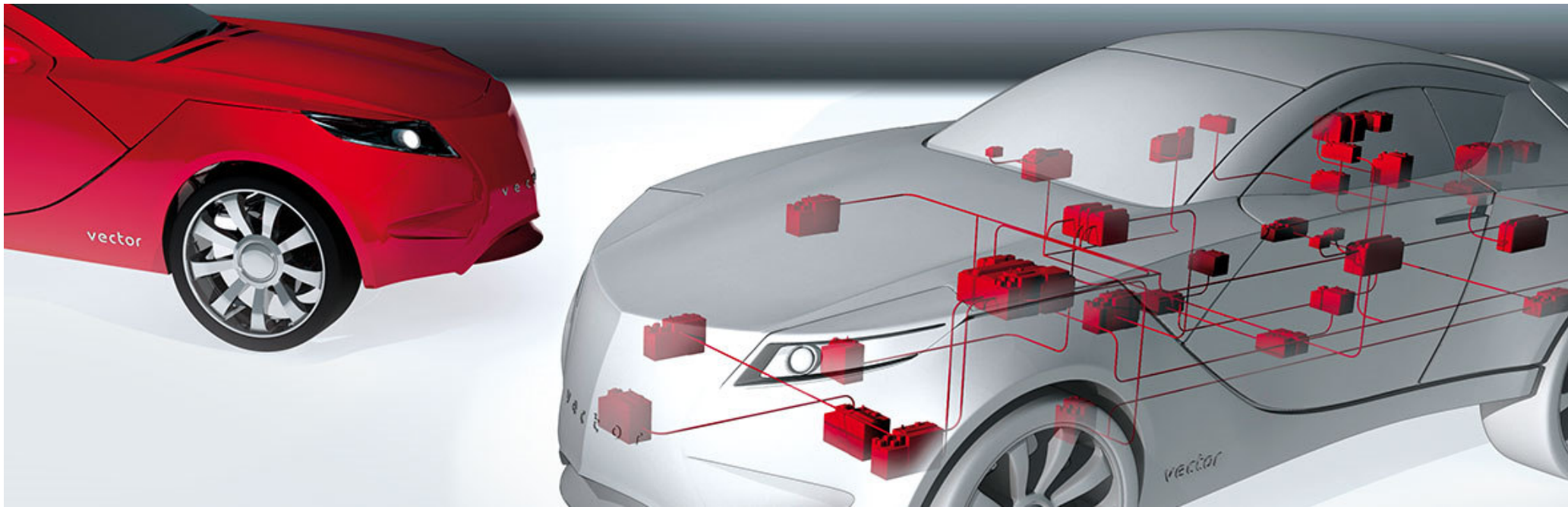




Overview on Automotive Ethernet

Vector India Conference 2019

Agenda

1.

Introduction

2.

Who Are The Actors

3.

Technology

4.

Ethernet vs. Traditional Bus Systems

5.

Protocols And Their Applications

Wired Acronyms And Terms

- Simple Question! What are you doing?

802.3bv

GEPOF

Ethernet

RTPGE

IEEE

802.3bw

802.3bp

1TPCE

1000BASE-T1

BroadR-Reach

OABR

IP

OPEN

100BASE-T1

Wired Acronyms And Terms – Let's Sort The Terms

- ▶ IEEE 802.3 Task Force names

GEPOF

Ethernet

802.3bv

RTPGE

IEEE

802.3bp

802.3bw

1TPCE

1000BASE-T1

BroadR-Reach

- ▶ 1TPCE = One (**1**) **T**wisted **P**air 100 Megabit (**C** = century = 100) **E**thernet
- ▶ RTPGE = **R**educed **T**wisted **P**air **G**igabit **E**thernet
- ▶ GEPOF = **G**igabit Ethernet **O**ver **P**lastic **O**ptical **F**iber

IP

QABR

OPEN

100BASE-T1

Wired Acronyms And Terms – Let's Sort The Terms

- ▶ IEEE 802.3xx → Name of a project within an IEEE Working Group

802.3bv

GEPOF

Ethernet

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BroadR-Reach

- ▶ bw = One (**1**) **T**wisted **P**air 100 Megabit (**C** = century = 100) **E**thernet
- ▶ bp = **R**educed **T**wisted **P**air **G**igabit **E**thernet
- ▶ bv = **G**igabit Ethernet **O**ver **P**lastic **O**ptical **F**iber

IP

QABR

OPEN

100BASE-T1

Wired Acronyms And Terms – Let's Sort The Terms

► Name of the physical layer

- 100BASE-T1 = 100 Megabit Baseband One Pair
- 1000BASE-T1 = 1 Gigabit Baseband One Pair
- 1000BASE-RH = Gigabit Ethernet Over Plastic Optical Fiber

1000BASE-T1

100BASE-T1

Wired Acronyms And Terms – Let's Sort The Terms

► Organizations and Technology

- OABR, (OPEN Alliance) BroadR-Reach
 - Early name for 100BASE-T1 when IEEE has not been involved and OPEN Alliance introduced Broadcom's BroadR-Reach technology to the automotive world.
 - OPEN, OPEN Alliance = One Pair Ethernet Network Alliance

IEEE

802.3bw

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OPEN

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Organizations



One Pair Ethernet Network Alliance

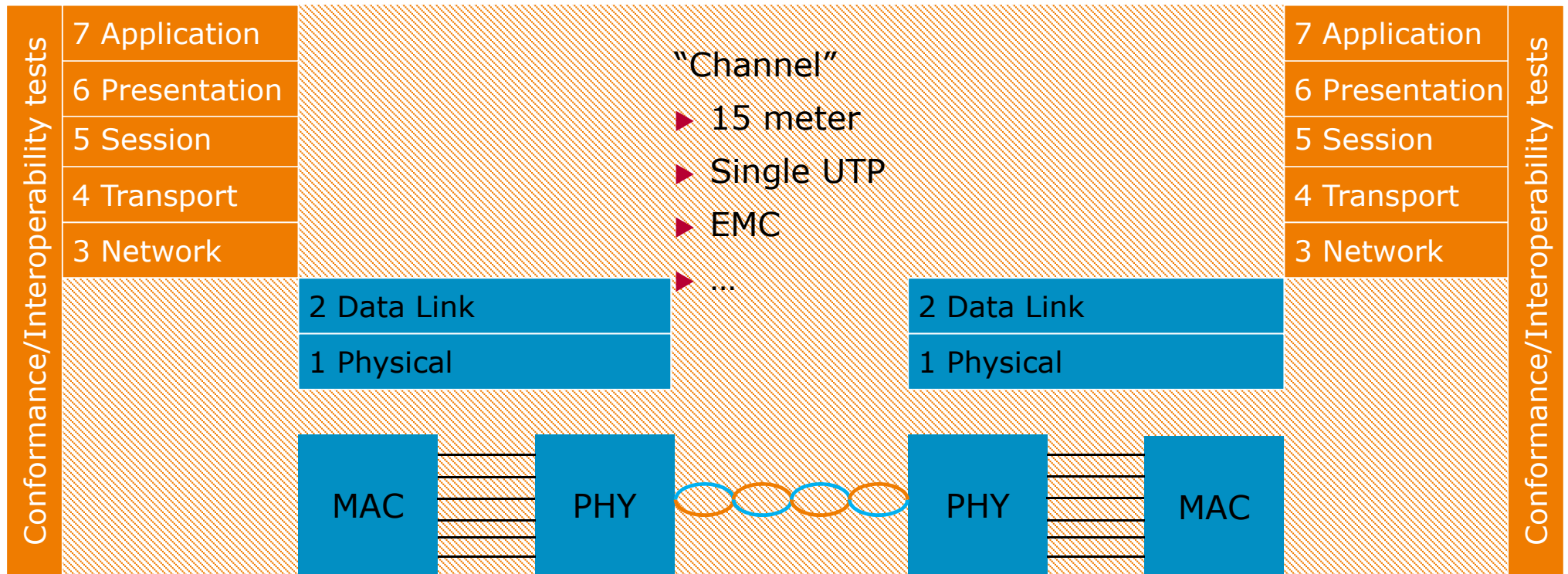
- ▶ "One voice" from the automobile industry
- ▶ Two types of membership (companies)
 - ▶ Promoters (16)
 - ▶ Adopters (359)
- ▶ Hosts 14 Technical Committees (TC)
 - ▶ Address all open items, not harmonized within other documents. E.g. compliance tests, interoperability tests, minimum requirements, ...



Institute of Electrical and Electronics Engineers

- ▶ 802.3 „Home" of Ethernet

Separate Areas of Activity



ISO/IEC

- ▶ OPEN Alliance hands over all documents to ISO for long term availability
- ▶ OPEN Alliance still creates/maintains the documents
- ▶ ISO standard is ISO 21111 Road vehicles – In-vehicle Ethernet and its parts
 - ▶ Part 1*** : General information and definitions
 - ▶ Part 2***: System requirements and physical layer interfaces
 - ▶ Part 3**: Optical 1-Gbit/s physical layer
 - ▶ Part 4**: General requirements and test methods of optical gigabit ethernet components
 - ▶ Part 5***: Optical 1-Gbit/s physical layer system specification and interoperability test plan
 - ▶ Part 6*: Electrical 100-Mbit/s physical layer device specification and conformance test plan
 - ▶ Part 7*: Electrical 100-Mbit/s physical layer system specification and interoperability test plan
 - ▶ Part 8*: Electrical 100-Mbit/s component requirements and test methods
 - ▶ Part 9*: Bridge specification and conformance test plan
 - ▶ Part 10*: General device requirements and test methods
 - ▶ Part 11****: Electrical 1-Gbit/s physical layer device specification and conformance test plan
 - ▶ Part 12****: Electrical 1-Gbit/s physical layer system specification and interoperability test plan
 - ▶ Part 13****: Electrical 1-Gbit/s component requirements and test methods

*New Project; ** Committee Draft; *** Approved New Work Item; **** Preliminary Work Item

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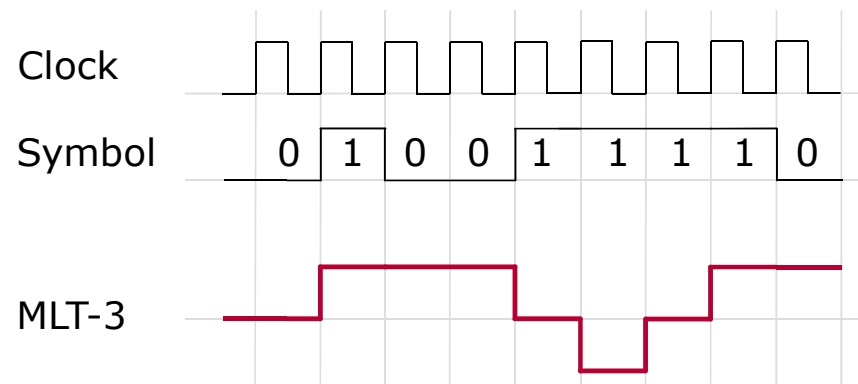
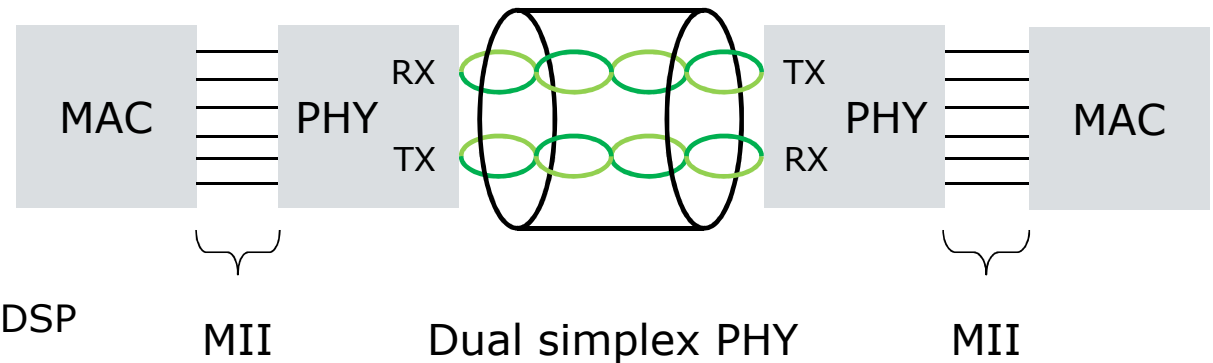
Ethernet vs. Traditional Bus Systems

5.

Protocols And Their Applications

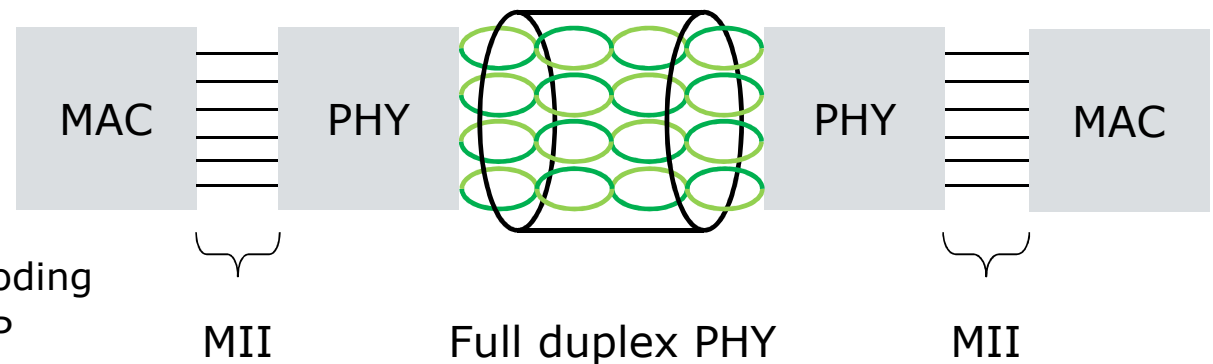
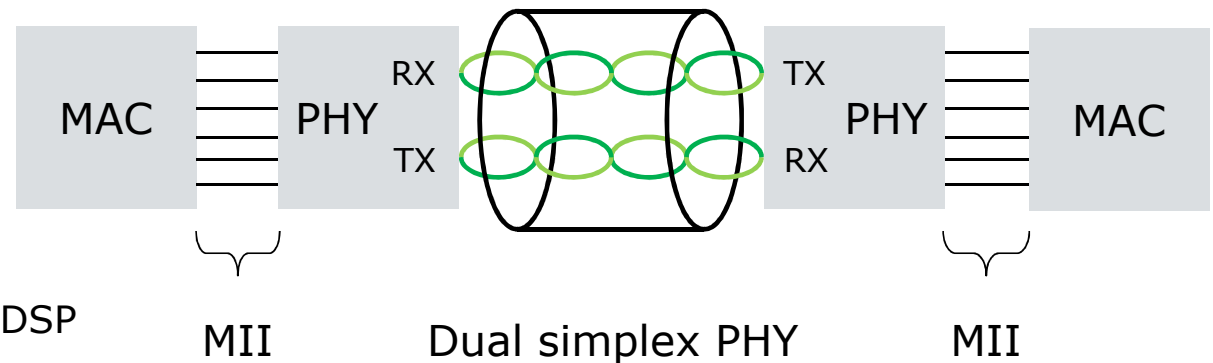
What is 100BASE-T1

- ▶ Combination of existing technologies!
- ▶ IEEE 100BASE-TX
 - ▶ Dual simplex
 - ▶ MLT-3 (Multi Level Transmission)
 - > 125 Msps, 65~80 MHz bandwidth
 - ▶ Two twisted pairs
 - ▶ No error correction coding
 - ▶ No Echo and crosstalk cancelation in DSP
 - ▶ Decision Feedback Equalization (DFE)



What is 100BASE-T1

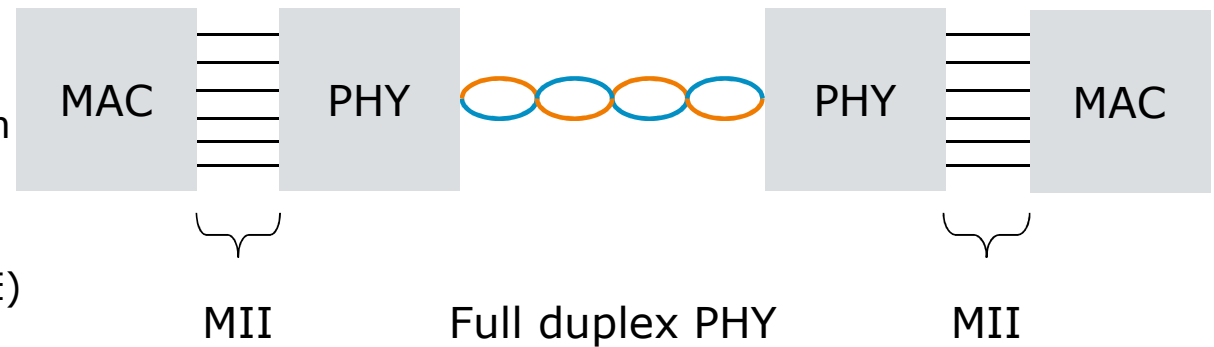
- ▶ Combination of existing technologies!
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 - ▶ Dual Simplex
 - ▶ MLT-3 (Multi Level Transmission)
 - > 125 Msps, 65~80 MHz bandwidth
 - ▶ Two twisted pairs
 - ▶ No error correction coding
 - ▶ No Echo and crosstalk cancelation in DSP
 - ▶ Decision Feedback Equalization (DFE)
- ▶ IEEE 1000BASE-T
 - ▶ Full Duplex
 - ▶ 4D-PAM5
 - > 125 Msps, 65~80 MHz bandwidth
 - ▶ Four twisted pairs
 - ▶ Partial response transmit filter
 - ▶ Additional level for error correction coding
 - ▶ Echo and crosstalk cancelation in DSP
 - ▶ Decision Feedback Equalization (DFE)



What is 100BASE-T1

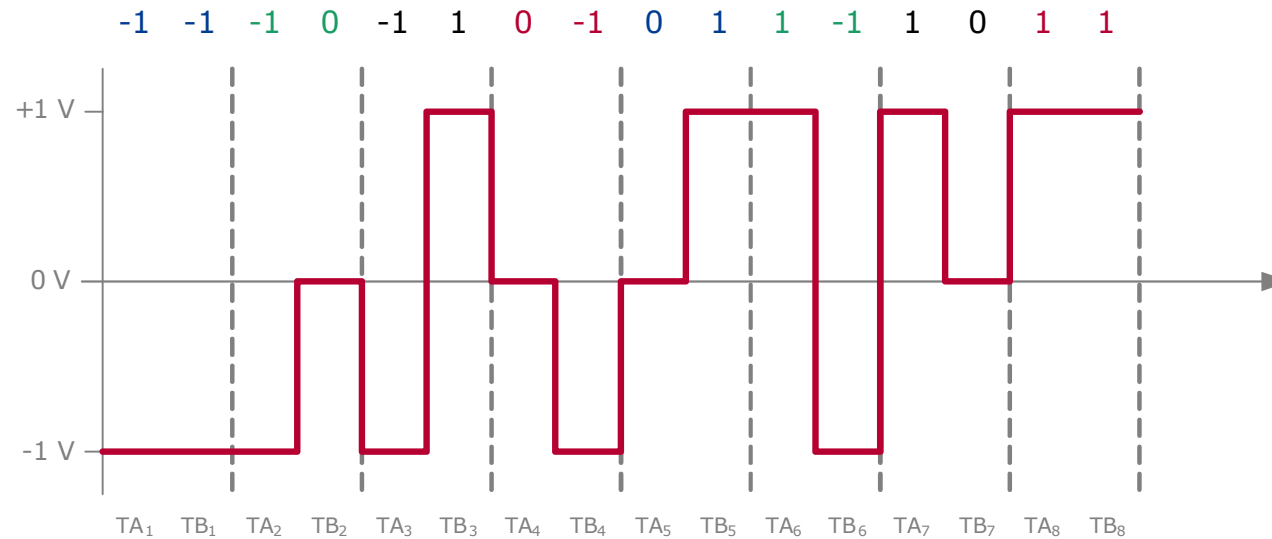
► 100BASE-T1

- Full Duplex
- PAM3, 66.7 Msps, 27 MHz bandwidth
- Single twisted pairs
- Echo cancelation
- Decision Feedback Equalization (DFE)



3B2T:

PAM3:



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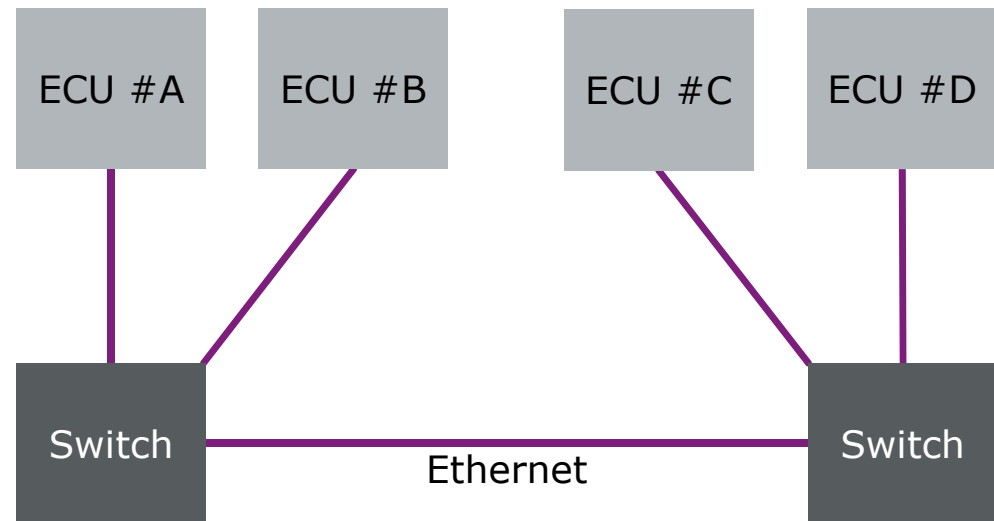
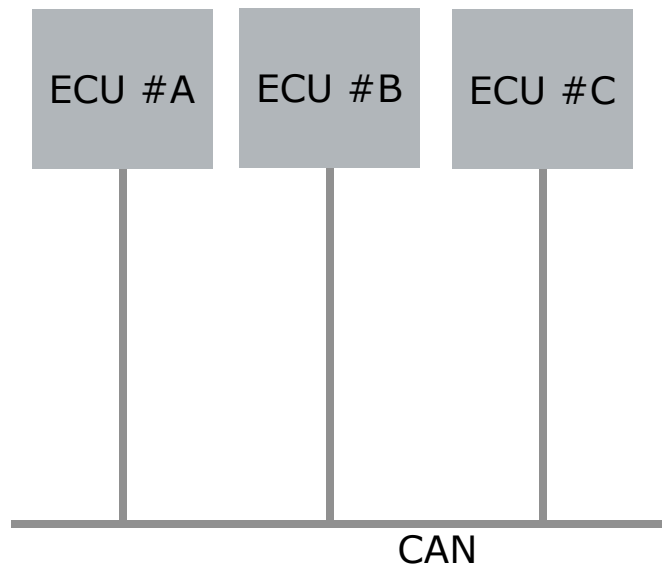
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Ethernet vs. Traditional Bus Systems

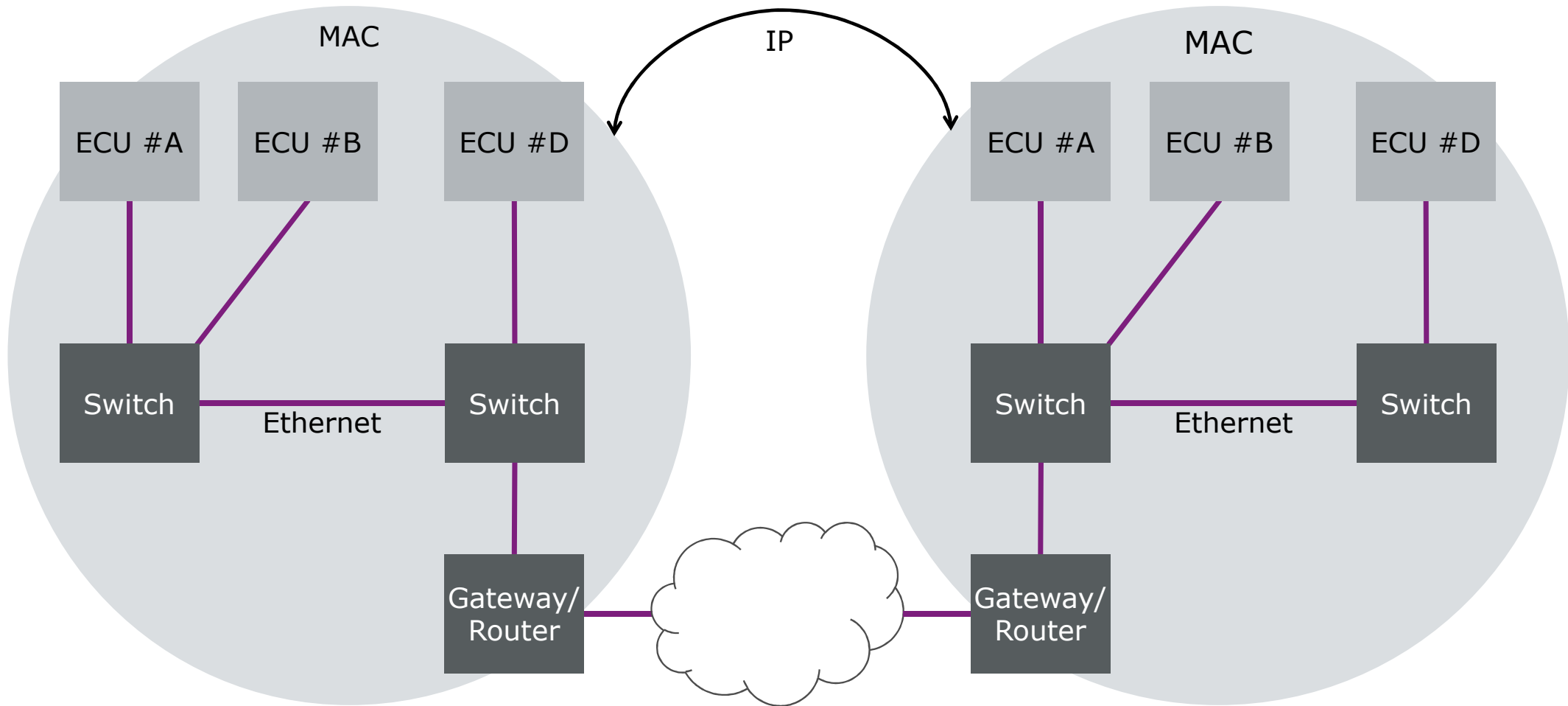
5.

Protocols And Their Applications

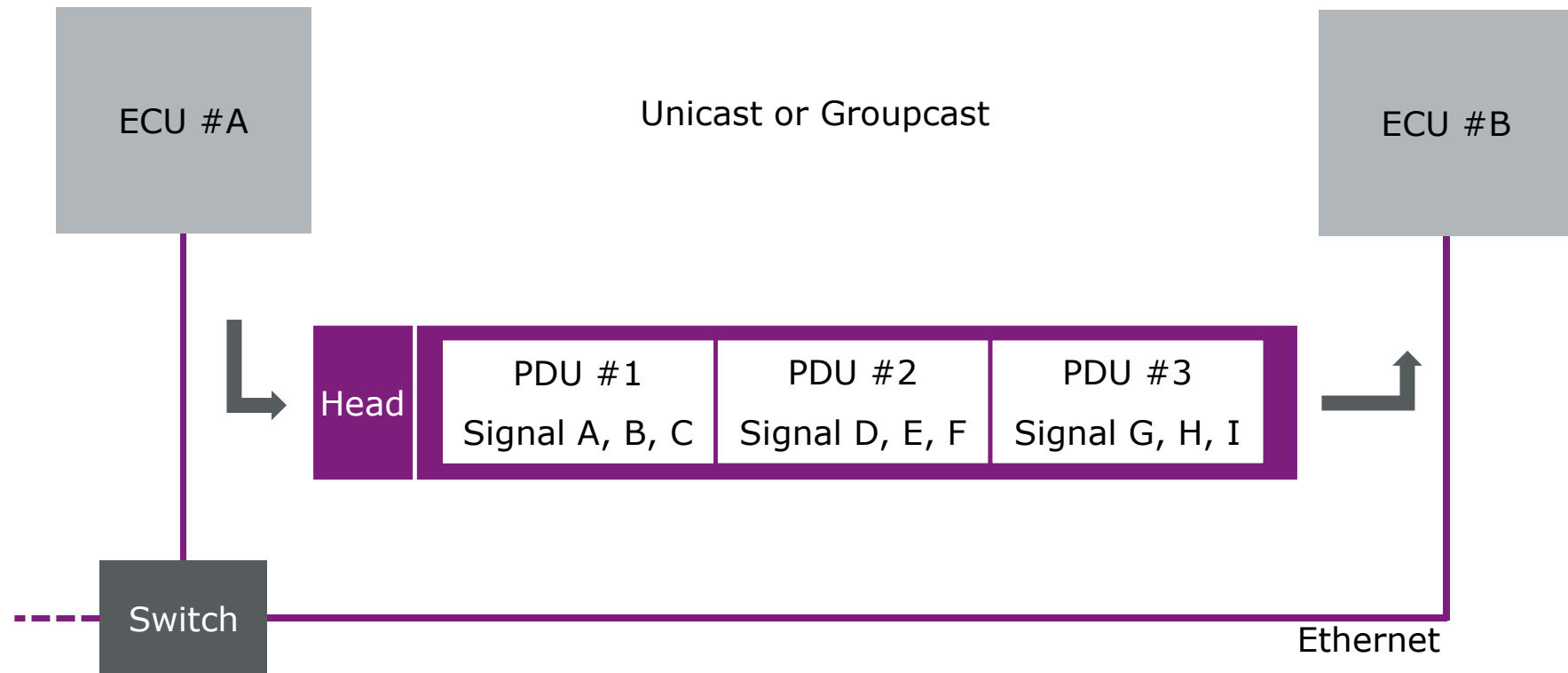
Switched Networks



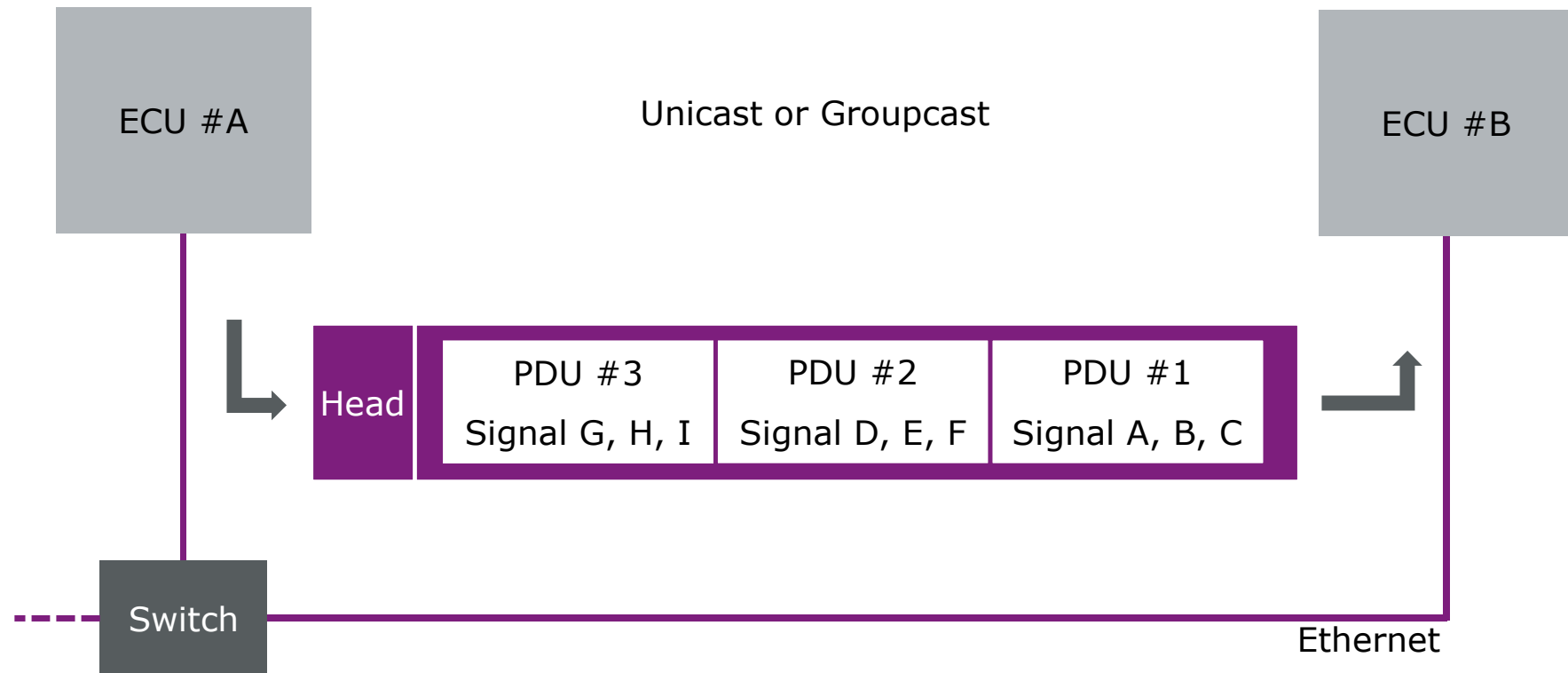
Connecting Networks – IP vs. MAC



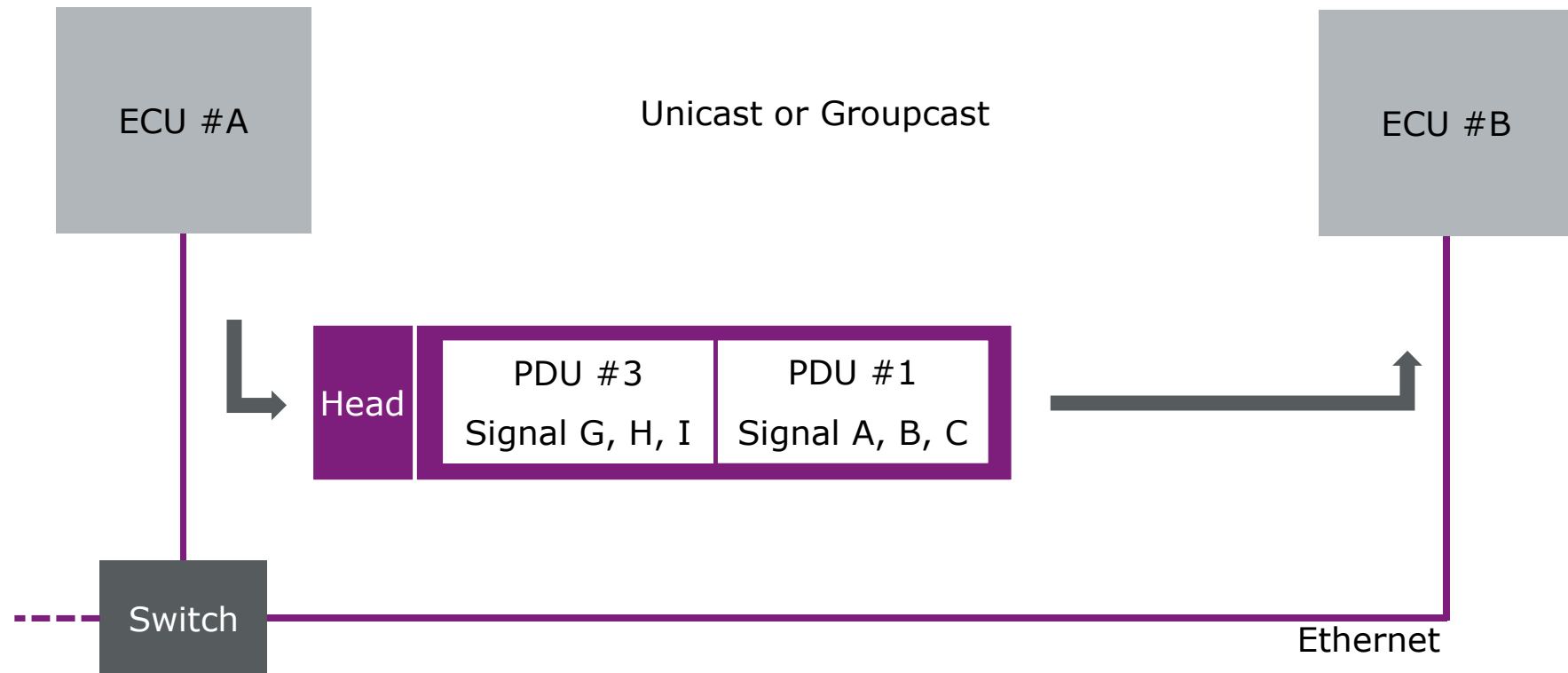
Signals And PDUs (Protocol Data Unit)



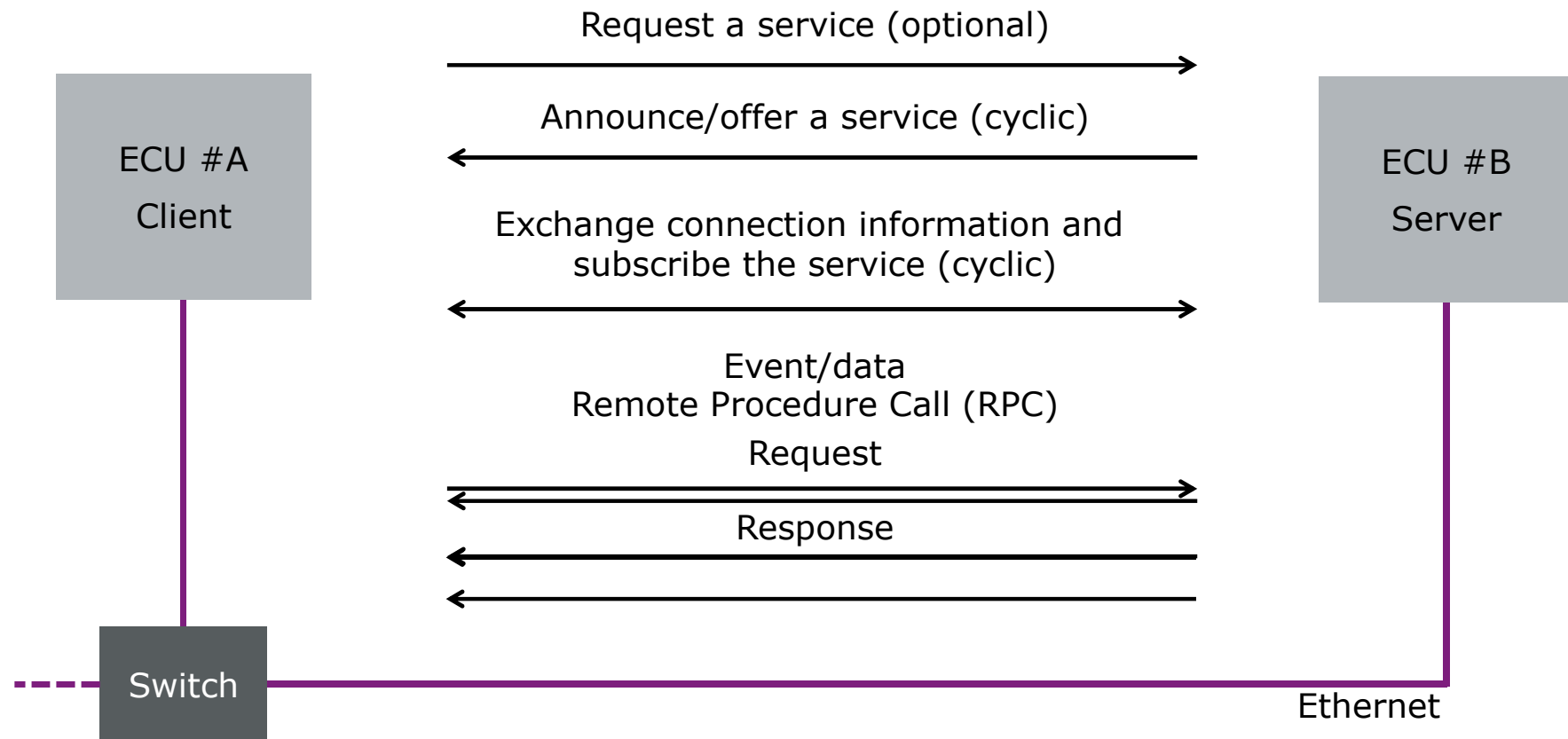
Signals And PDUs (Protocol Data Unit)



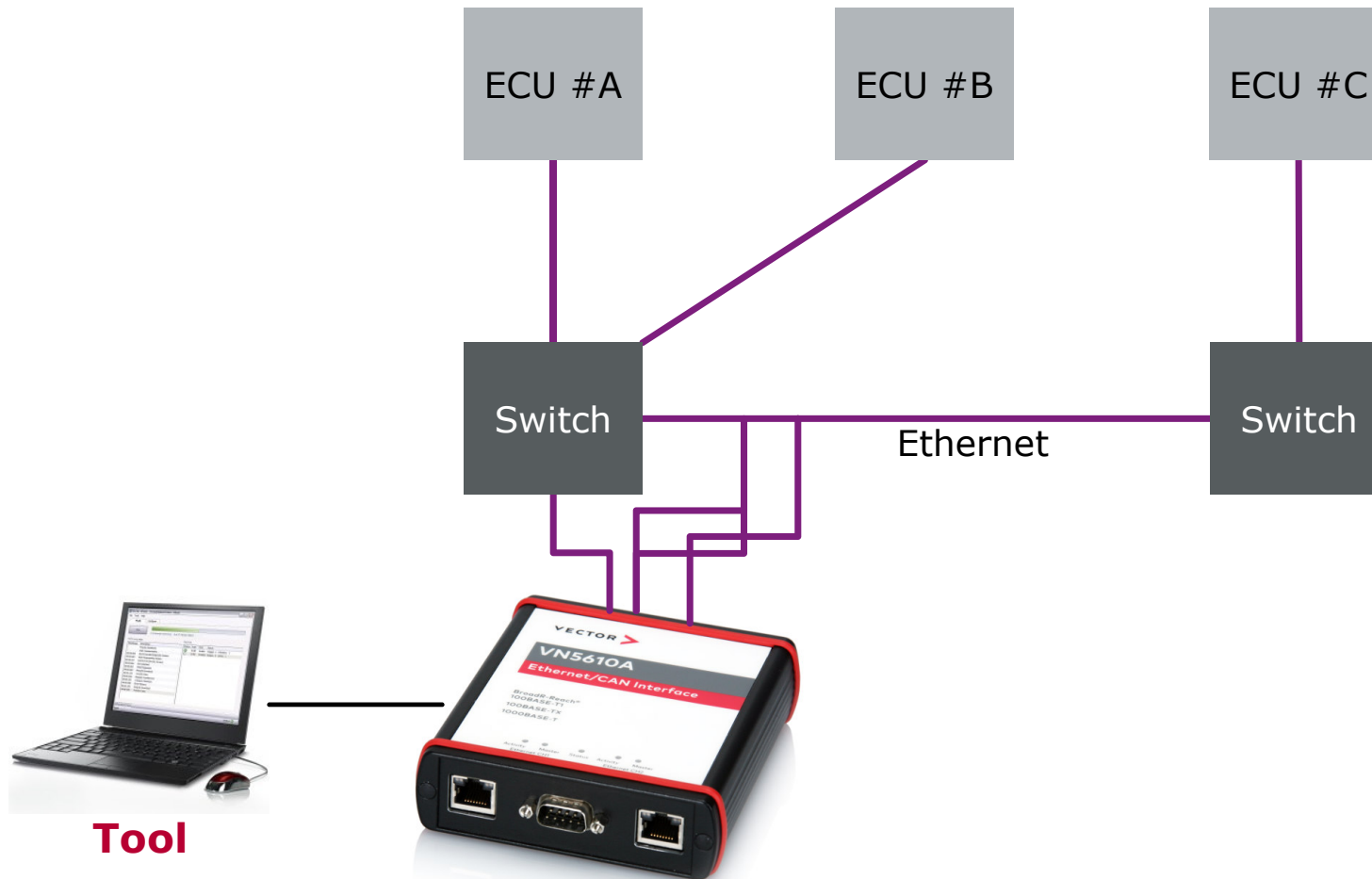
Signals And PDUs (Protocol Data Unit)



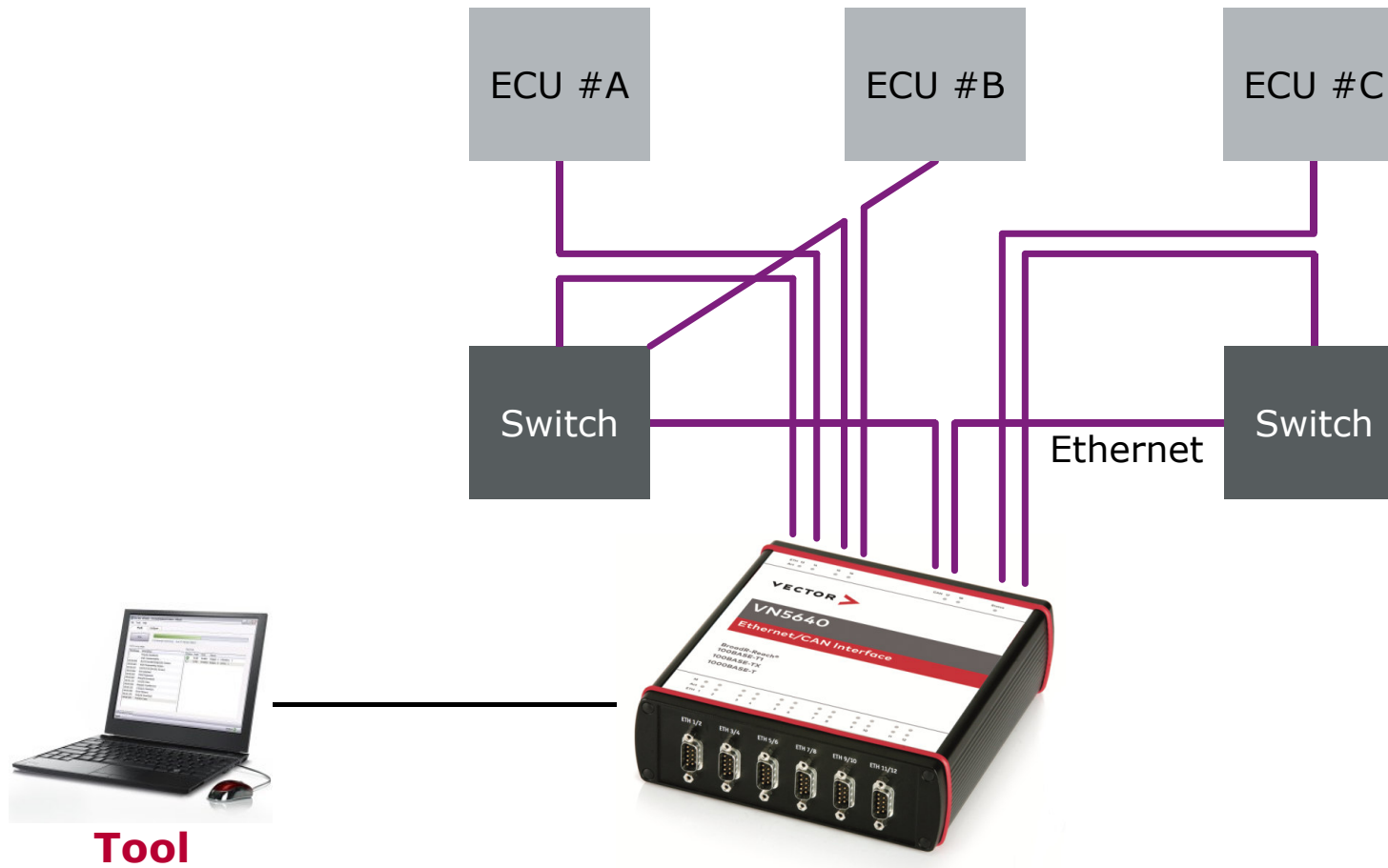
Service Oriented Communication and Service Discovery



How to Access The Network?

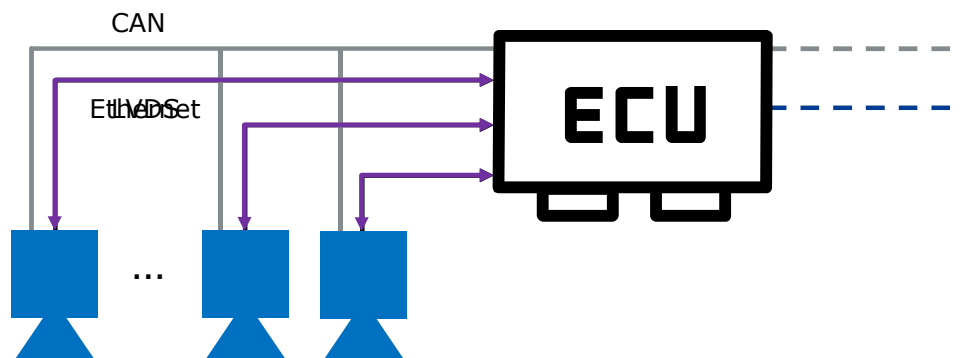
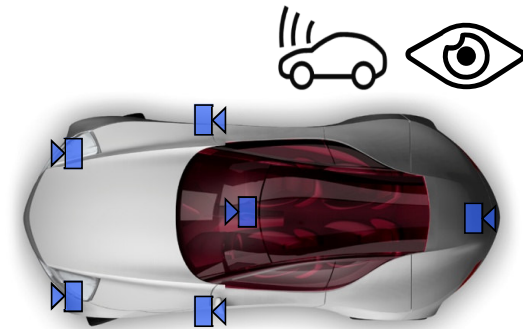


How to Access The Network?

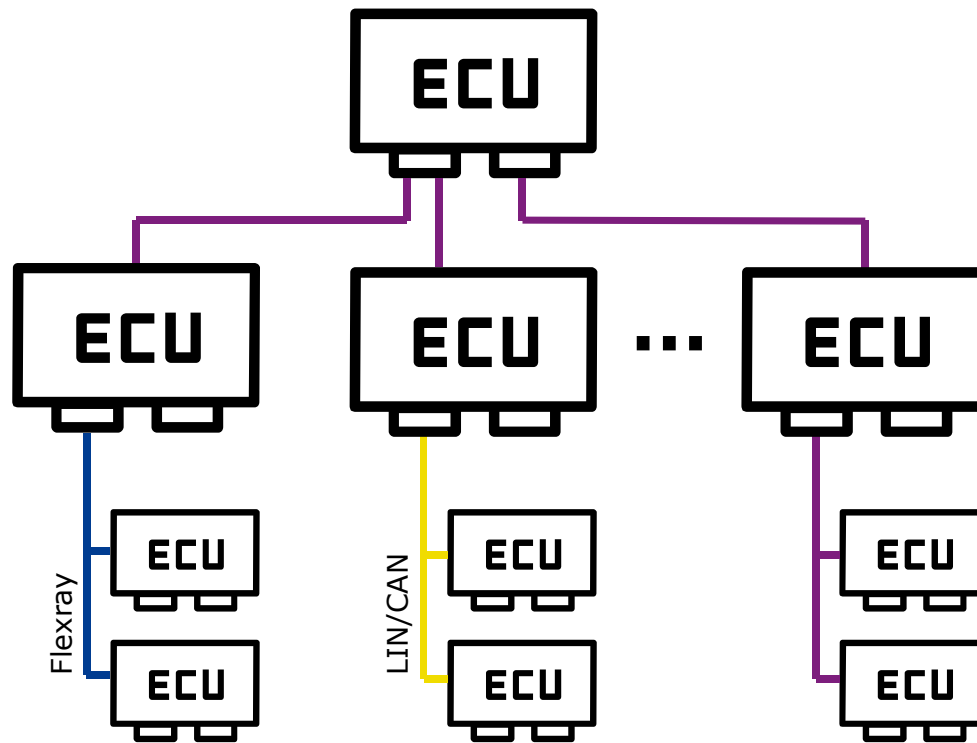


Bird View – High Speed Link

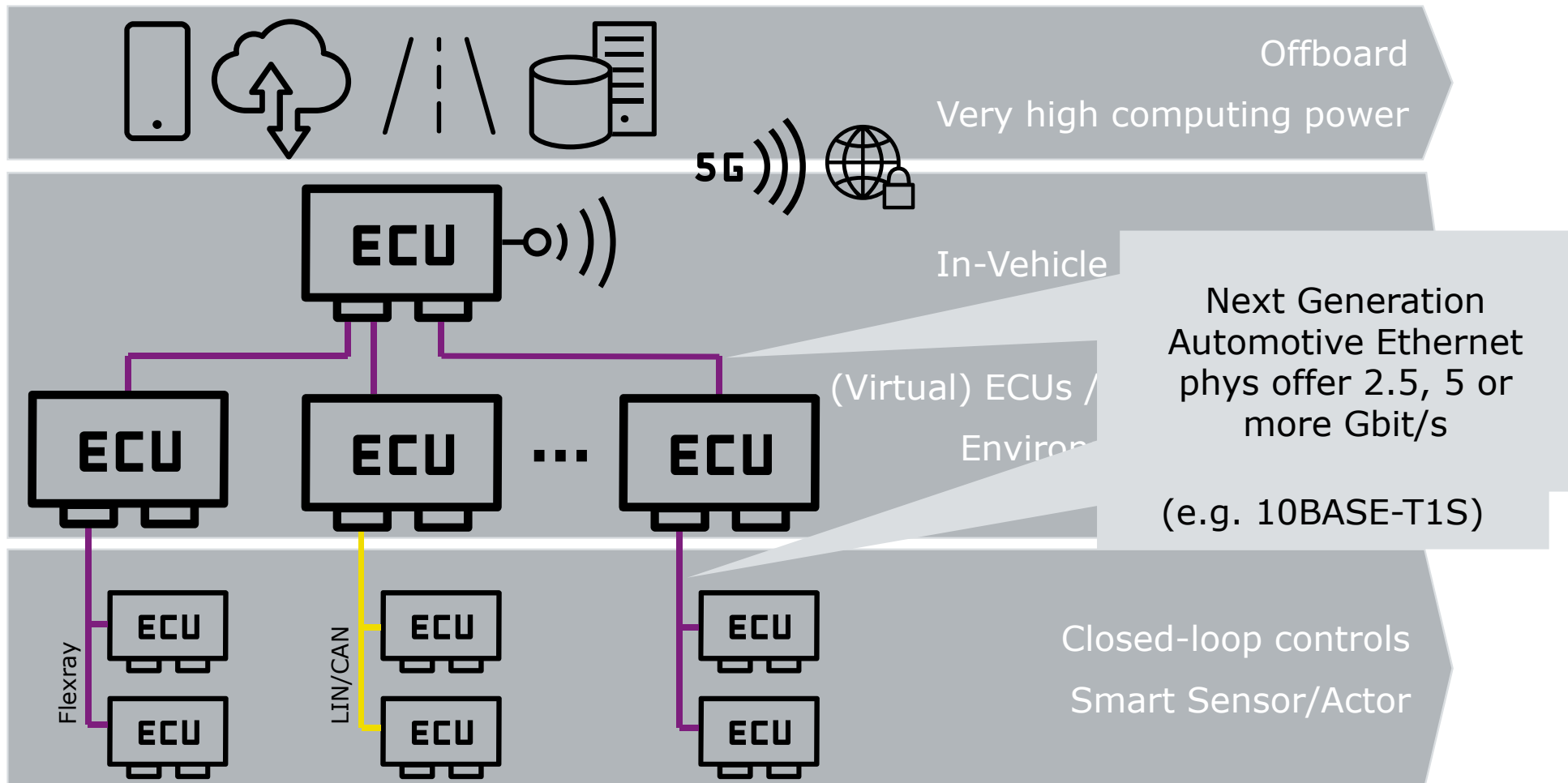
- ▶ Broadcom BroadR-Reach, 100 Mbit/s, full duplex, twisted pair
 - ▶ Bird View has been Day-One application for 100BASE-T1
 - ▶ Replace expensive shielded cable
 - ▶ Bi-Directional communication
- ▶ Utilizing high bandwidth
- ▶ MAC based Video streaming, time critical
- ▶ Enhanced by ADAS functionality (additional cameras, radar, lidar, ...)
 - ▶ High resolution and frame rate, less compression
 - ▶ Requires 1000 Mbit/s and more



Backbone – Connect Domain Controllers



Infrastructure – Connect Application Server



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